Chapter 6: Database Design Using the E-R Model

Database Principle and Design

Outline

- Overview of the Design Process
- The Entity-Relationship Model
- Complex Attributes
- Mapping Cardinalities
- Primary Key
- Removing Redundant Attributes in Entity Sets
- Reducing E-R Diagrams to Relational Schemas

6.1 Design Phases

- Initial phase -- fully describe the data requirements of future database users.
- Second phase (conceptual-design phase)
  - Translating the data requirements into a conceptual schema of the database. This schema represents overall logical structure of the database.
- Final Phase -- Moving from a conceptual schema to the implementation of the database
  - Logical Design – Deciding on the database schema.
  - Physical Design – Deciding on the physical layout of the database

User Requirements → Conceptual Schema → Database Schema

Pitfalls in Design

- In designing a database schema, we must ensure that we avoid two major pitfalls:
  - Redundancy: a bad design may result in repeat information that may lead to data inconsistency among the various copies of information.
    - For example, in the section relation, we repeat all course information once for each section of the course
  - Incompleteness: a bad design may make certain aspects of the enterprise difficult or impossible to model.
    - For example, we have only a *section* relation, but not a *course* relation. It would then be impossible to represent information about a new course, unless that course is offered.

course_id	sec_id	semest...	year	building	room_num...	time_slot...	title	dept_name	credits
BIO-101	1	Summer	2017	Painter	514	B	Intro. to Biology	Biology	4
BIO-301	1	Summer	2018	Painter	514	A	Genetics	Biology	4
CS-101	1	Fall	2017	Packard	101	H	Intro. to Computer Science	Comp. Sci.	4
CS-101	1	Spring	2018	Packard	101	F	Intro. to Computer Science	Comp. Sci.	4
CS-190	1	Spring	2017	Taylor	3128	E	Game Design	Comp. Sci.	4
CS-190	2	Spring	2017	Taylor	3128	A	Game Design	Comp. Sci.	4
CS-315	1	Spring	2018	Watson	120	D	Robotics	Comp. Sci.	3
CS-319	1	Spring	2018	Watson	100	B	Image Processing	Comp. Sci.	3
CS-319	2	Spring	2018	Taylor	3128	C	Image Processing	Comp. Sci.	3
CS-347	1	Fall	2017	Taylor	3128	A	Database System Concepts	Comp. Sci.	3

Data Redundancy:

6.2 The Entity-Relationship (E-R) Model

- The **E-R data model** was developed to facilitate database design by allowing specification of a **conceptual schema**.
- The **E-R data model** models an enterprise as a collection of *entities* and *relationships.*
  - Entity: a "thing" or "object" in the enterprise that is distinguishable from other objects
    - Described by a set of *attributes*
  - Relationship: an association among several entities
- The E-R model also has an associated diagrammatic representation, the **E-R diagram**, which can express the overall logical structure of a database graphically.

Entity Sets

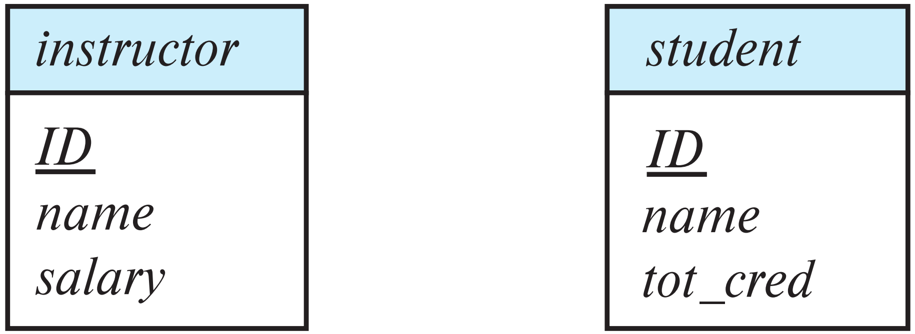
- An **entity** is an object that exists and is distinguishable from other objects.
  - Example: student *Mary*, instructor *Smith*, course *History*, flight *NH301*
- An **entity set** is a set of entities of the same type that share the same properties.
  - Example: *Students, Instructors, Courses, Flights*
- An entity is represented by a set of attributes; i.e., descriptive properties possessed by all members of an entity set.
  - Example:

    instructor = (*ID, name, salary*)
    course= (*course_id, title, credits*)
- A subset of the attributes form a **primary key** of the entity set; i.e., uniquely identifying each member of the set.

Note: We only include entities and attributes that are useful to the system in the E-R model

Representing Entity Sets in E-R Diagram

- Entity sets can be represented graphically as follows:
  - Rectangles represent entity sets.
  - Attributes listed inside entity rectangle
  - Underline indicates primary key attributes



Note: *Both instructor and student omit the dept_name attribute because it represents the relationship with the department. In E-R model, relationships between entity sets should be represented explicitly by relationship set, not implicitly by an attribute in entity set*

Relationship Sets

- A **relationship** is an association among several entities

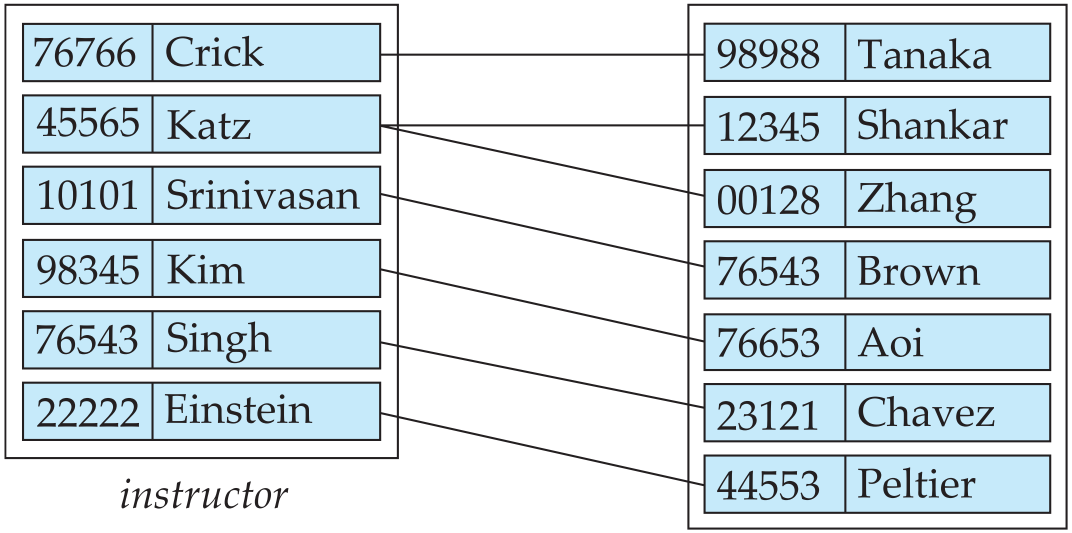
  Example:

22222 (Einstein)	*advisor*	44553 (Peter)
instructor entity	relationship	*student* entity

- A **relationship set** is a set of relationships of the same type.
- A **relationship set** R is a mathematical relation on $n \geq 2$ entity sets (E_1, E_2, …, E_n):

  $$\{(e_1, e_2, \ldots e_n) \mid e_1 \in E_1, e_2 \in E_2, \ldots, e_n \in E_n\}$$

  where (e_1, e_2, …, e_n) is a relationship



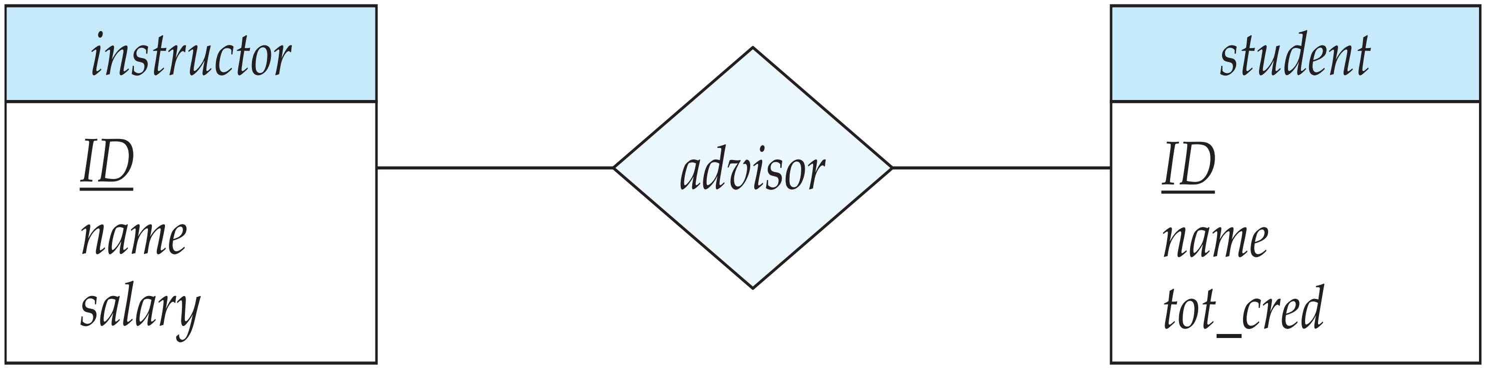
instructor

advisor relationship set

student

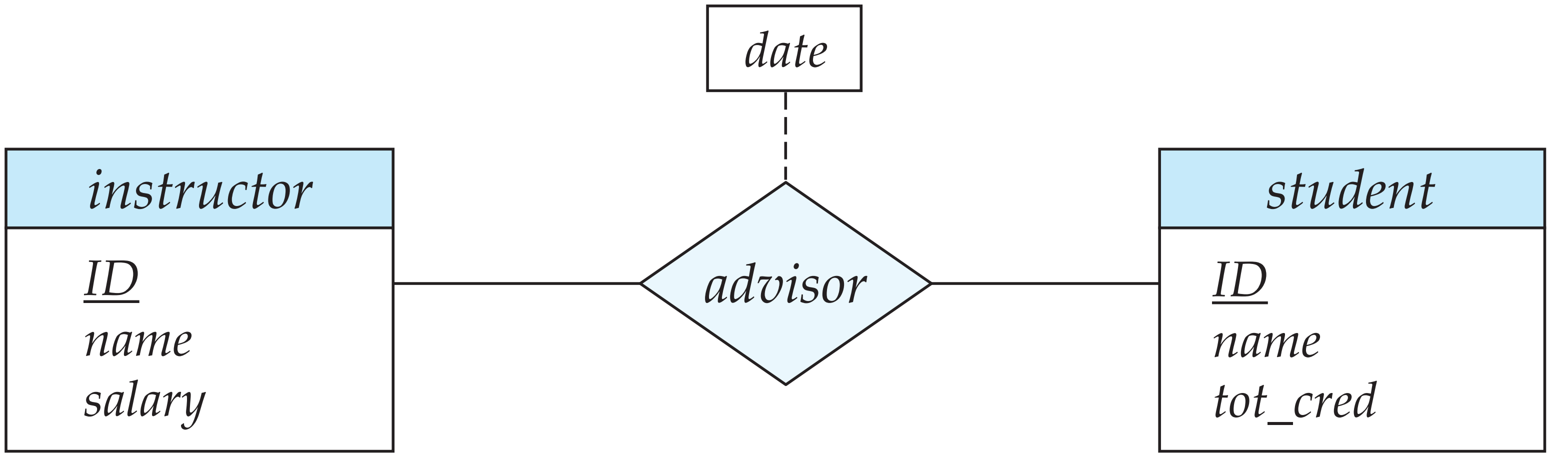
Representing Relationship Sets in E-R Diagrams

- A relationship set is represented by a diamond, which is linked via lines to the entity sets.



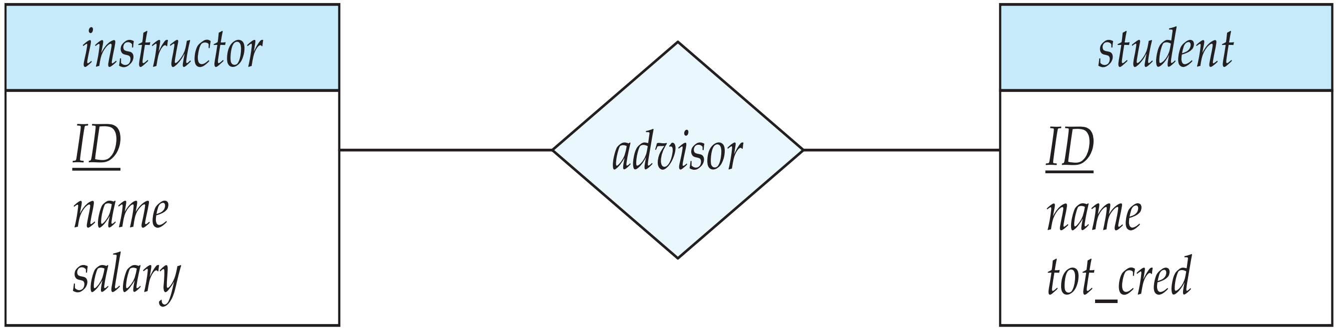
Relationship Sets with Attributes

- A relationship may also have attributes called **descriptive attributes**.
- For instance, the *advisor* relationship set between entity sets *instructor* and *student* may have a descriptive attribute *date* which tracks when the student started being associated with the advisor

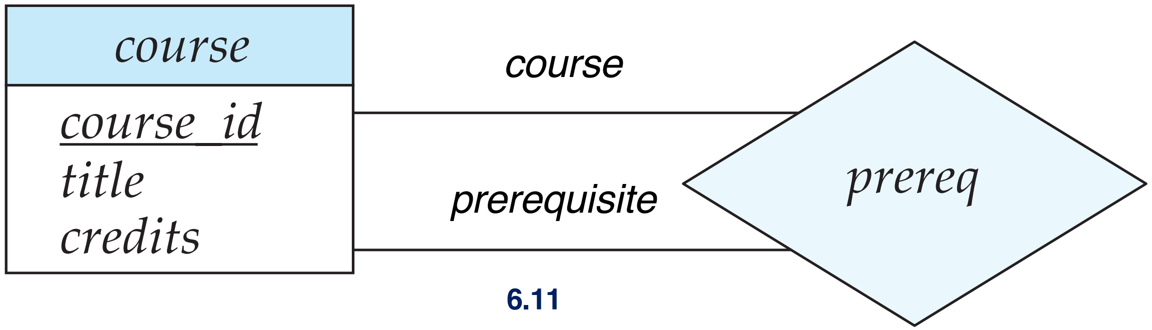


Roles

- Each occurrence of an entity set plays a "role" in the relationship set.
- Since entity sets participating in a relationship set are generally distinct, roles are implicit and are not usually specified.

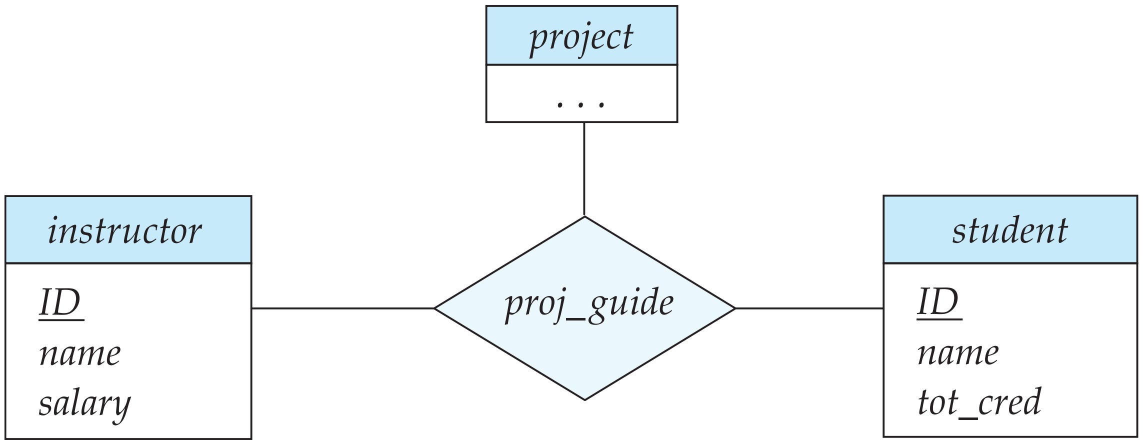


- Roles need to be specified when the same entity set participates in a relationship set more than once, in different roles. For example: *prereq* relationship set between courses.
  - Roles are indicated in E-R diagrams by labeling the lines that connect diamonds to rectangles



Degree of a Relationship Set

- The number of entity sets that participate in a relationship set is the degree of the relationship set.
- Most relationship sets are **binary**.
- Relationships between more than two entity sets are rare.
  - Example: relationship *proj_guide* is a **ternary** relationship between *instructor, student,* and *project,* representing a student working on a projects under the guidance of an instructor.

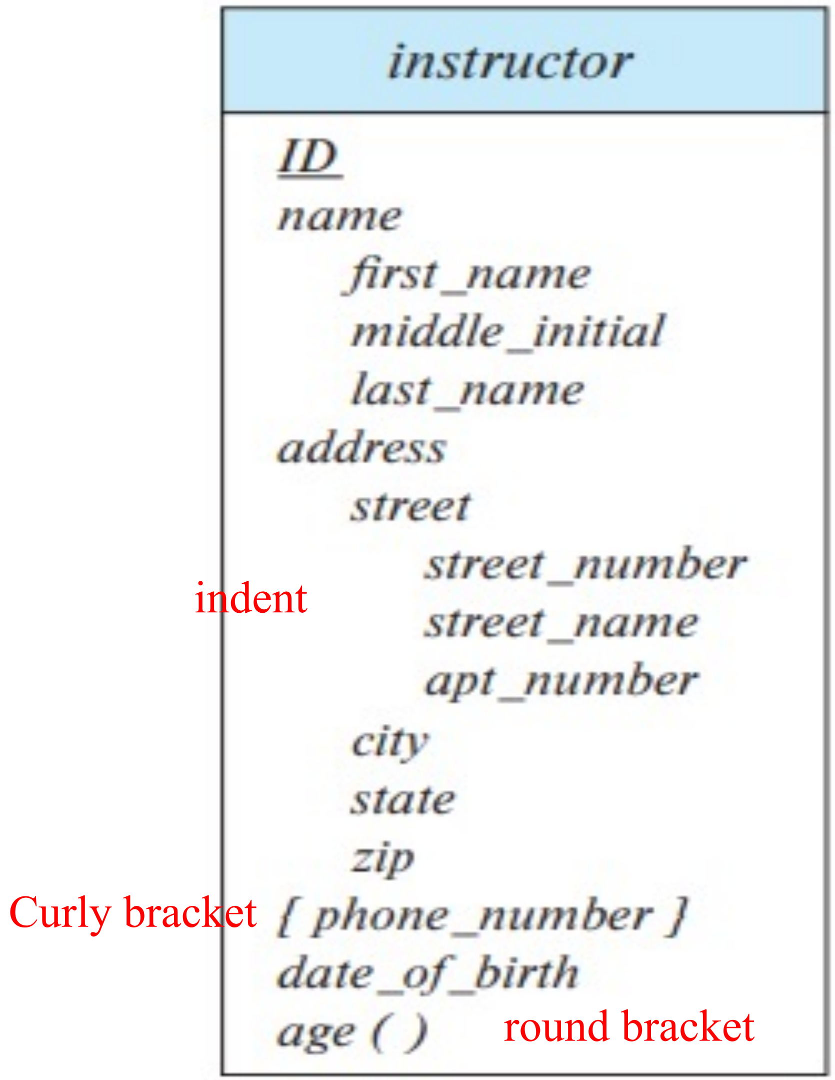


Note: This course only considers binary relationships.

6.3 Complex Attributes

- **Multivalued attribute**: its value contains multiple values of same type. For example: *phone_numbers*
- **Composite attribute**: its value can be divided into subparts. For example: the *address* attribute consists of street, city, state, and postal code.
- **Derived** attributes
  - Can be computed from other attributes
  - Example: date_of_birth -> age

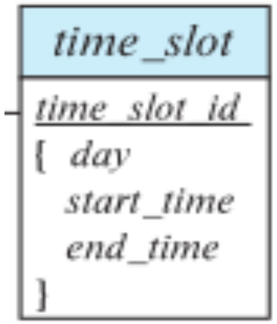
Representing Complex Attributes in E-R Diagram



Example: Time slot

- *Section time is allocated by time slots. A time slot contains multiple time intervals, each interval contains 3 items.*

course_id	sec_id	semester	year	building	room_number	time_slot_id	day	start_time	end_time
BIO-101	1	Summer	2017	Painter	514	B	F	9	10
BIO-101	1	Summer	2017	Painter	514	B	M	9	10
BIO-101	1	Summer	2017	Painter	514	B	W	9	10
BIO-301	1	Summer	2018	Painter	514	A	F	8	9
BIO-301	1	Summer	2018	Painter	514	A	M	8	9
BIO-301	1	Summer	2018	Painter	514	A	W	8	9
CS-101	1	Fall	2017	Packard	101	H	W	10	13
CS-101	1	Spring	2018	Packard	101	F	R	14	16
CS-101	1	Spring	2018	Packard	101	F	T	14	16
CS-190	1	Spring	2017	Taylor	3128	E	R	10	12
CS-190	1	Spring	2017	Taylor	3128	E	T	10	12
CS-190	2	Spring	2017	Taylor	3128	A	F	8	9

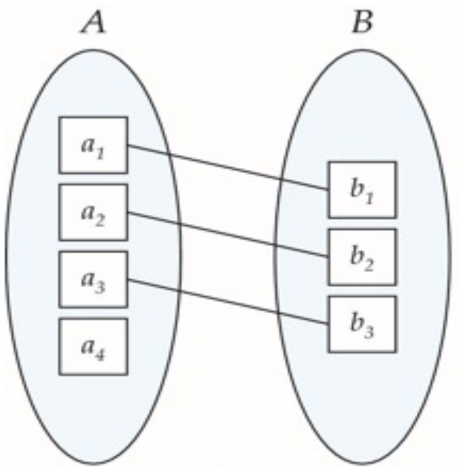


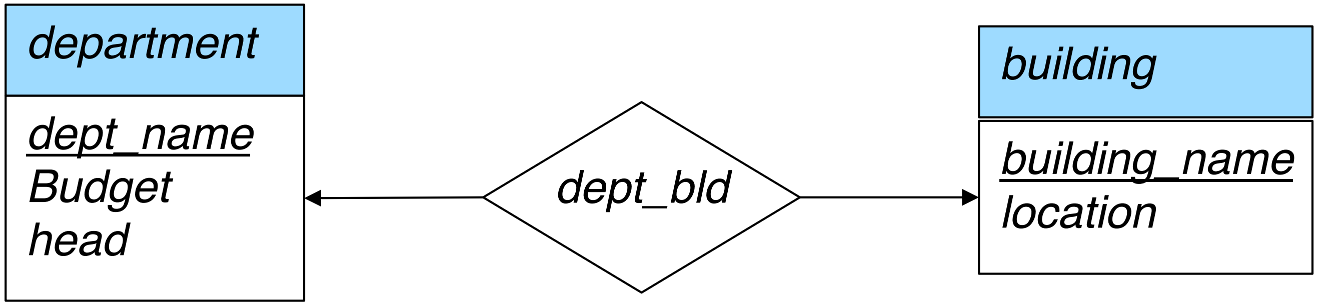
6.4 Mapping Cardinalities

- Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.
- It's mainly used in binary relationship sets.
- For a binary relationship set the mapping cardinality must be one of the following types:
  - One to one
  - One to many
  - Many to one
  - Many to many

Mapping Cardinalities (Cont.)

- **One-to-one**. An entity in A is associated with at most one entity in B, and an entity in B is associated with at most one entity in A.

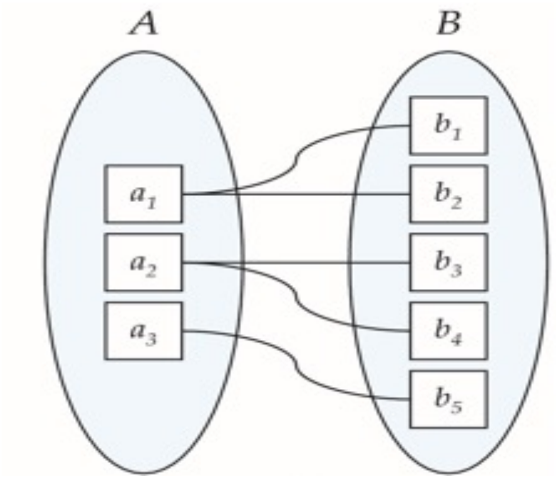




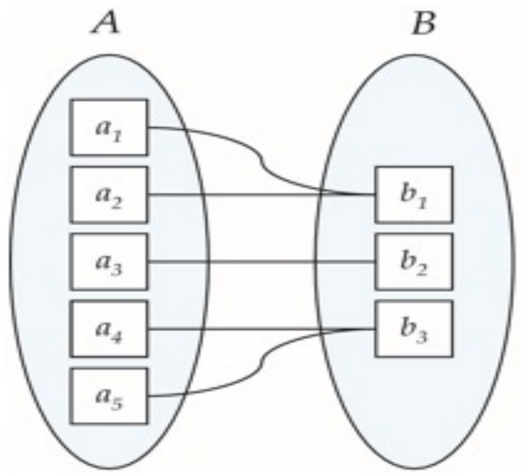
Note: Some elements in A and B may not be mapped to any elements in the other set

Mapping Cardinalities (Cont.)

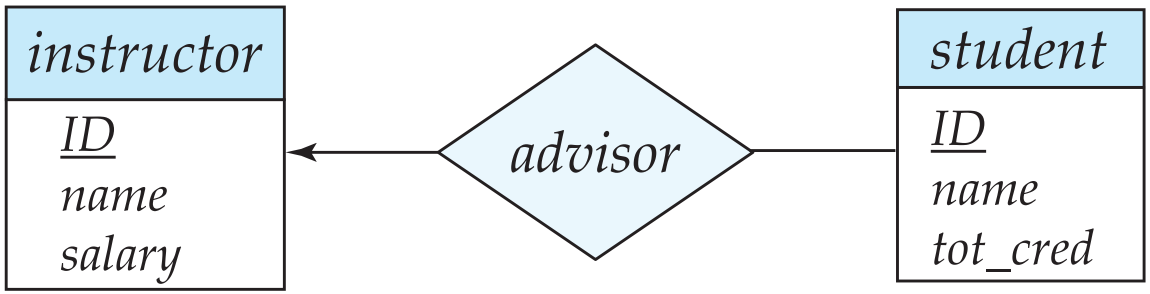
- **One-to-many**. An entity in A is associated with any number (zero or more) of entities in B. An entity in B, however, can be associated with at most one entity in A.
- **Many-to-one**. An entity in A is associated with at most one entity in B. An entity in B, however, can be associated with any number (zero or more) of entities in A.



One to many

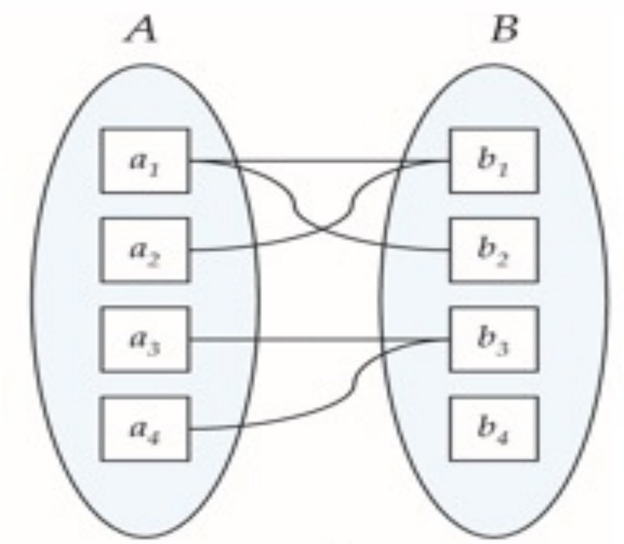


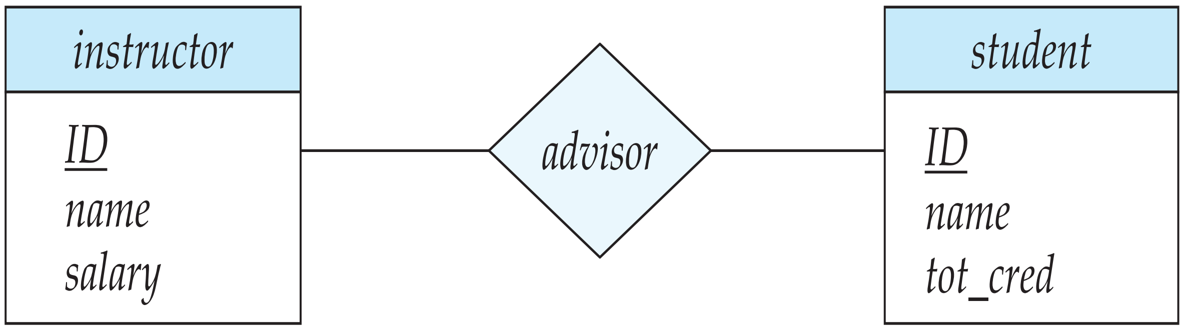
Many to one



Mapping Cardinalities (Cont.)

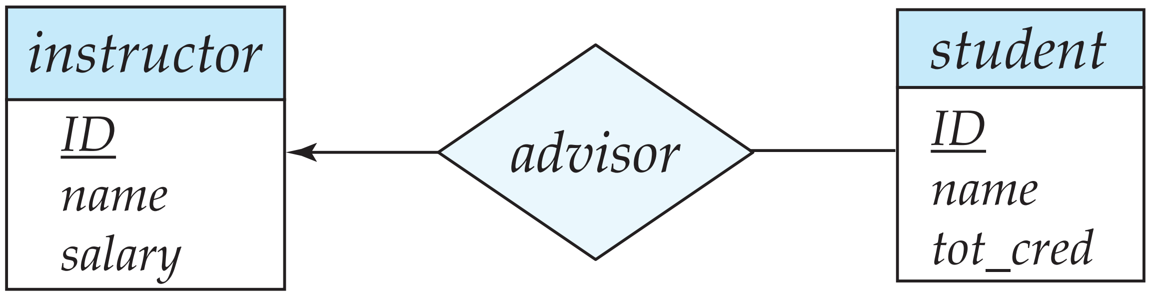
- **Many-to-many**. An entity in A is associated with any number (zero or more) of entities in B, and an entity in B is associated with any number (zero or more) of entities in A.



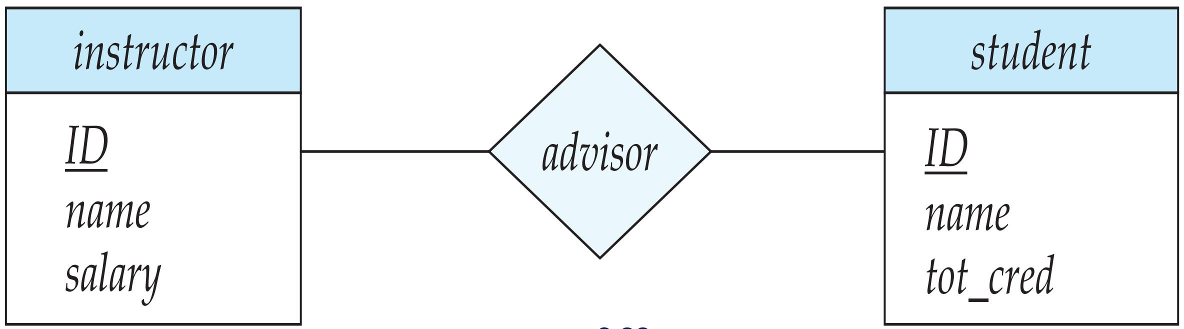


Representing Mapping Cardinality in E-R Diagram

- We express mapping cardinalities by drawing either a directed line (→), signifying "one," or an undirected line (—), signifying "many," between the relationship set and the entity set.
- Example:
  - one-to-many relationship between an *instructor* and a *student*

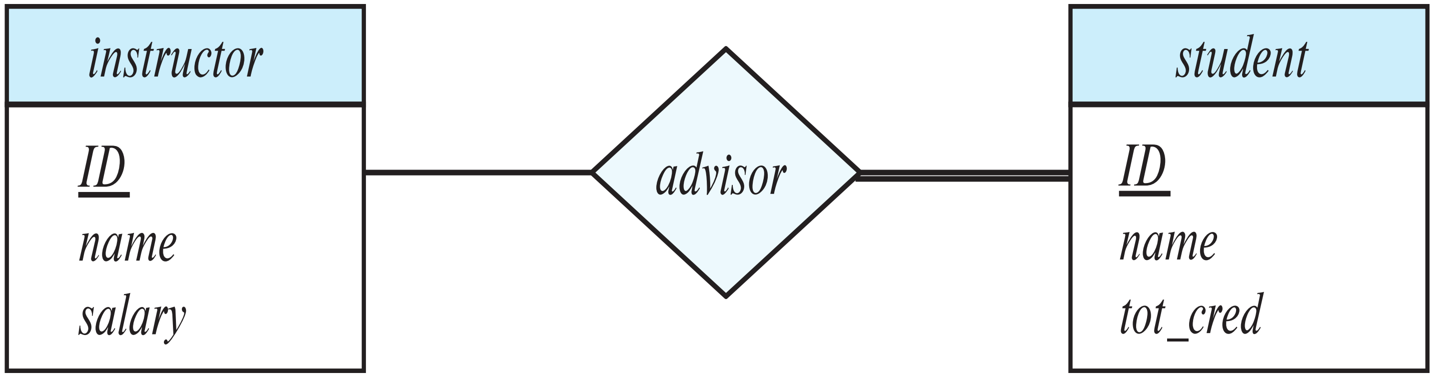


  - many-to-many relationship between an *instructor* and a *student*



Representing Participation of an entity set in E-R Diagram

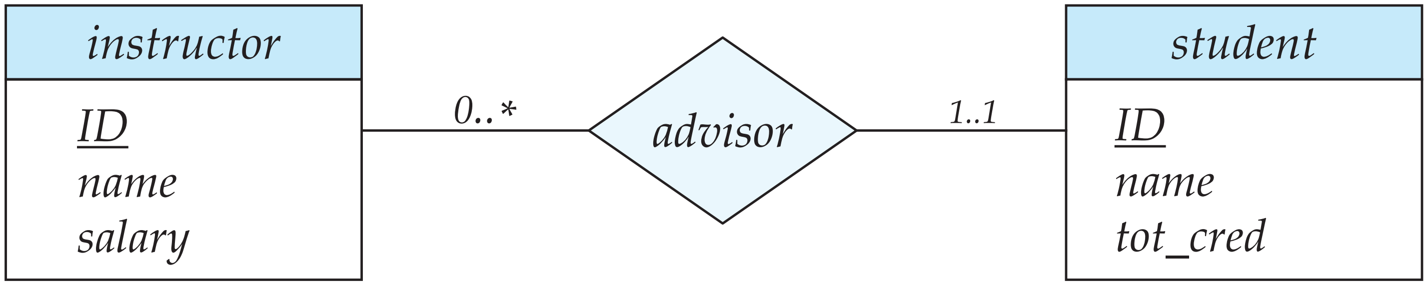
- The participation of an entity set E in a relationship set R:
  - **Total participation**: every entity in the entity set E participates in at least one relationship in the relationship set R



    - Example: participation of student in advisor relationship set is total
  - **Partial participation**: some entities may not participate in any relationship in the relationship set
    - Example: participation of *instructor* in *advisor* is partial
  - In E-R diagrams, total participations are represented by double lines, partial participations are represented by single lines.

Representing Cardinality Limits in E-R Diagram

- **Cardinality limits** express the minimum and maximum number of times **each** entity participates in a relationship set.
- It is shown in the form *l..h*, where *l* is the minimum and *h* the maximum cardinality
  - A minimum value of 1 indicates total participation.
  - A maximum value of 1 indicates that the entity participates in at most one relationship
  - A maximum value of * indicates no limit.
- Example



  - Instructor can advise 0 or more students. A student must have 1 advisor; cannot have multiple advisors

Note: Cardinality limits are optional in some E-R diagrams

6.5 Primary Key

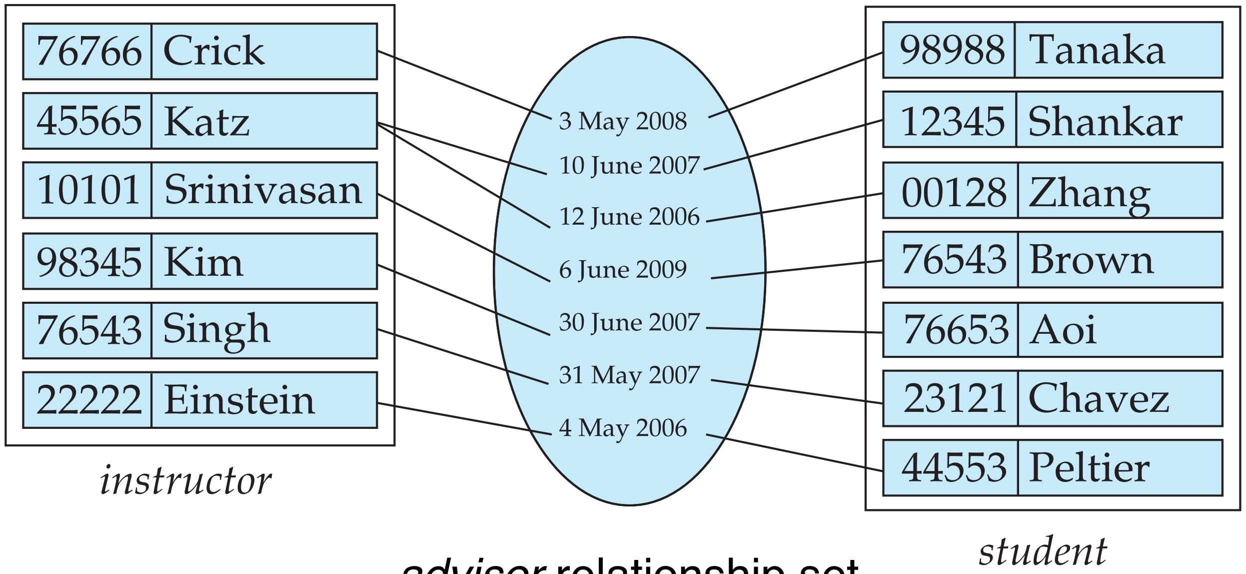
- In real world, individual entities and individual relationships are distinct.
- A key for an entity set or a relationship set is a set of attributes that suffice to distinguish entities or relationships from each other.
- The concepts of superkey, candidate key, and primary key are applicable to entity sets and relationship sets just as they are applicable to relation schemas.

Relation of Relationship Set

- A relationship set can be represented by a relation.
- Let *R* be a relationship set involving entity sets E1, E2, .. En, and has attributes a1, a2, .., am associated with it, then the set of attributes

  primary-key(E1) ∪ *primary-key(E2)* ∪ ··· ∪ *primary-key(En)* ∪ *{a1, a2,…, am}*

  forms a relation representing the relationship set *R*.



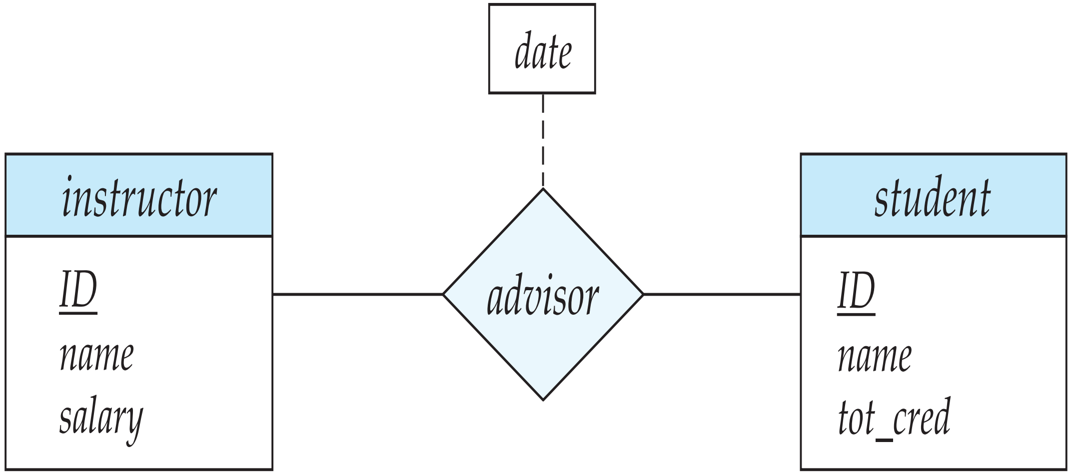
advisor relationship set

I_ID	S_ID	Date
76766	98988	2008-5-3
45565	12345	2007-6-10
45565	00128	2006-6-12
10101	76543	2009-6-6
98345	76653	2007-6-30
76543	23121	2007-5-31
22222	44553	2006-5-4

advisor relation

Primary Key for Relationship Sets

- The choice of the primary key for a relationship set depends on the mapping cardinality of the relationship set.
  - Many-to-Many relationship sets: The union of the **discriminator attributes** of the relationship set and the primary keys of the entity sets involved in the relationship set can be chosen as the primary key.



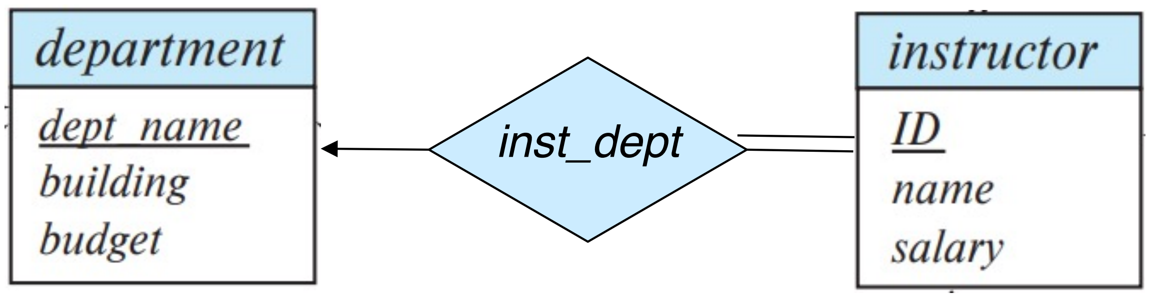
advisor relation

I_ID	S_ID	Date
76766	98988	2008-5-3
45565	98988	2007-6-10
76766	98988	2006-6-12
10101	76543	2009-6-6
….	….	….

Note: **Discriminator attributes** are those descriptive attributes involved in the primary key

Primary Key for Relationship Sets(Cont.)

- One-to-Many/ Many-to-one relationship sets . The primary key of the "Many" side can be used as the primary key.

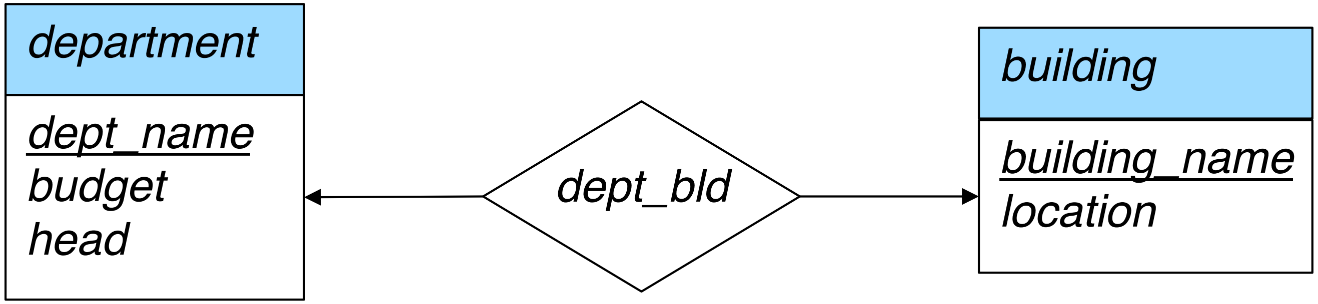


dept_name	ID
Comp. Sci.	10101
Finance	12121
Music	15151
Physics	22222
History	32343
Physics	33456
Comp. Sci.	45565
History	58583
Finance	76543
Biology	76766
Comp. Sci.	83821
Elec. Eng.	98345

inst_dept relation

Primary Key for Relationship Sets (Cont.)

- One-to-one relationship sets. The primary key of either one of the participating entity sets can be chosen as the primary key



dept_name	building_name
Biology	BIOLOGY-BUILDING
Comp. Sci.	COMP. SCI.-BUILDING
Elec. Eng.	ELEC. ENG.-BUILDING
Finance	FINANCE-BUILDING
History	HISTORY-BUILDING
Music	MUSIC-BUILDING
Physics	PHYSICS-BUILDING

dept_bld relation

Weak Entity Sets

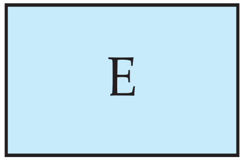
- A **weak entity set** is one whose existence is dependent on another entity set, called its **identifying entity set**. The identifying entity set is said to **own** the weak entity set that it identifies.
- For example, the *section* is a weak entity set whose existence is dependent on the *course* entity set.
- An entity set that is not a weak entity set is termed a **strong entity set**.
- The relationship associating the weak entity set with the identifying entity set is called the **identifying relationship**.
- Instead of associating a primary key with a weak entity, we use the primary key of the identifying entity, along with extra attributes called **discriminator attributes** to uniquely identify a weak entity.
- Primary key for *section* – (*course_id, sec_id, semester, year*)

Expressing Weak Entity Sets

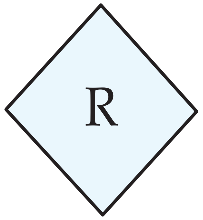
- In E-R diagrams, a weak entity set is depicted via a double rectangle.
- We underline the discriminator of a weak entity set with a dashed line.
- The relationship set connecting the weak entity set to the identifying strong entity set is depicted by a double diamond.

Note that we don't have the *course_id* attribute in the *section* entity set. The *course_id* is provided implicitly by identifying relationship.

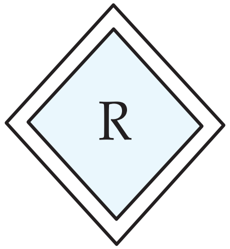
Summary of Symbols Used in E-R Notation



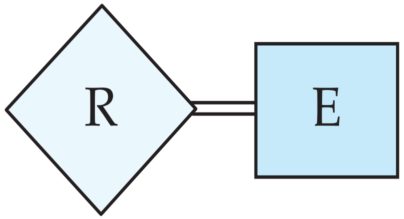
entity set



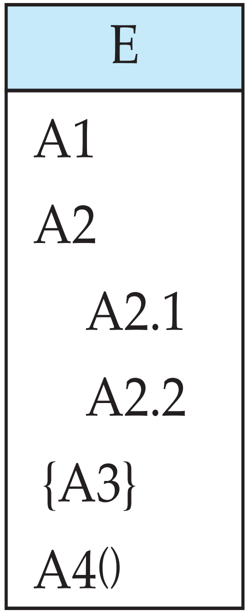
relationship set



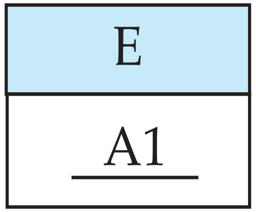
identifying relationship set for weak entity set



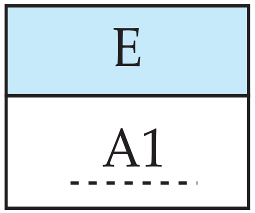
total participation of entity set in relationship



attributes:
simple (A1),
composite (A2) and
multivalued (A3)
derived (A4)

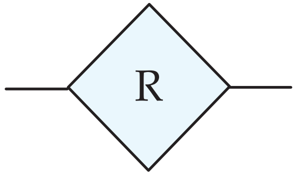


primary key

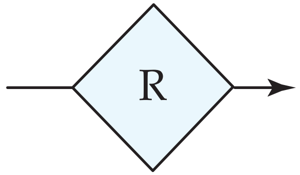


discriminating attribute of weak entity set

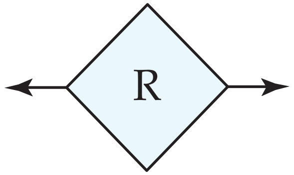
Symbols Used in E-R Notation (Cont.)



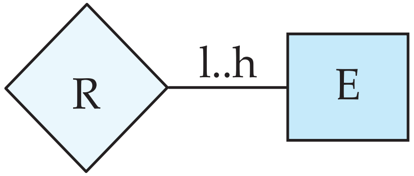
many-to-many relationship



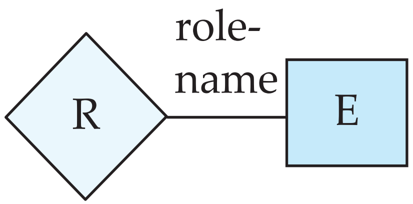
many-to-one relationship



one-to-one relationship

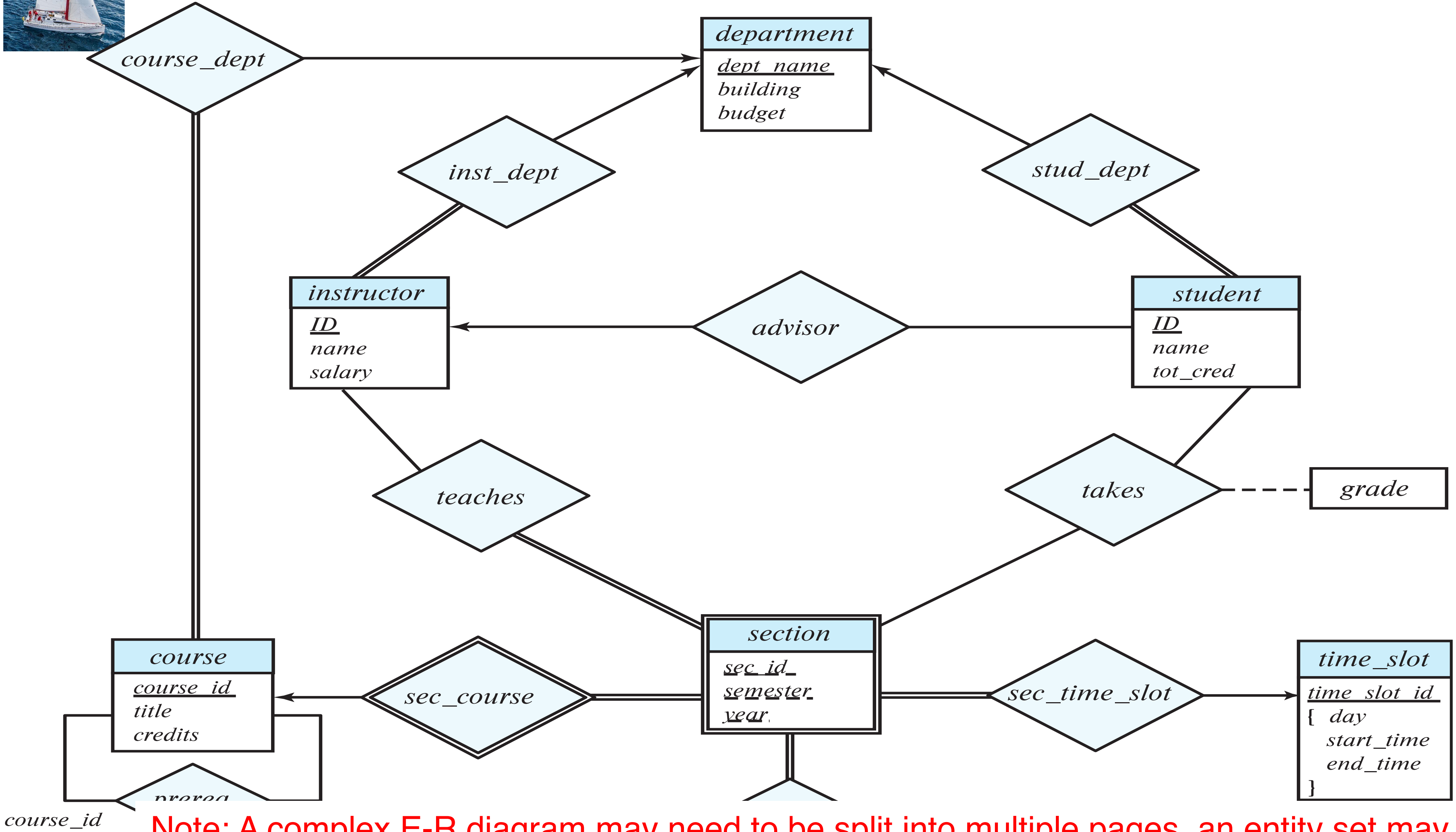


cardinality limits



role indicator

6.6 E-R Diagram for a University



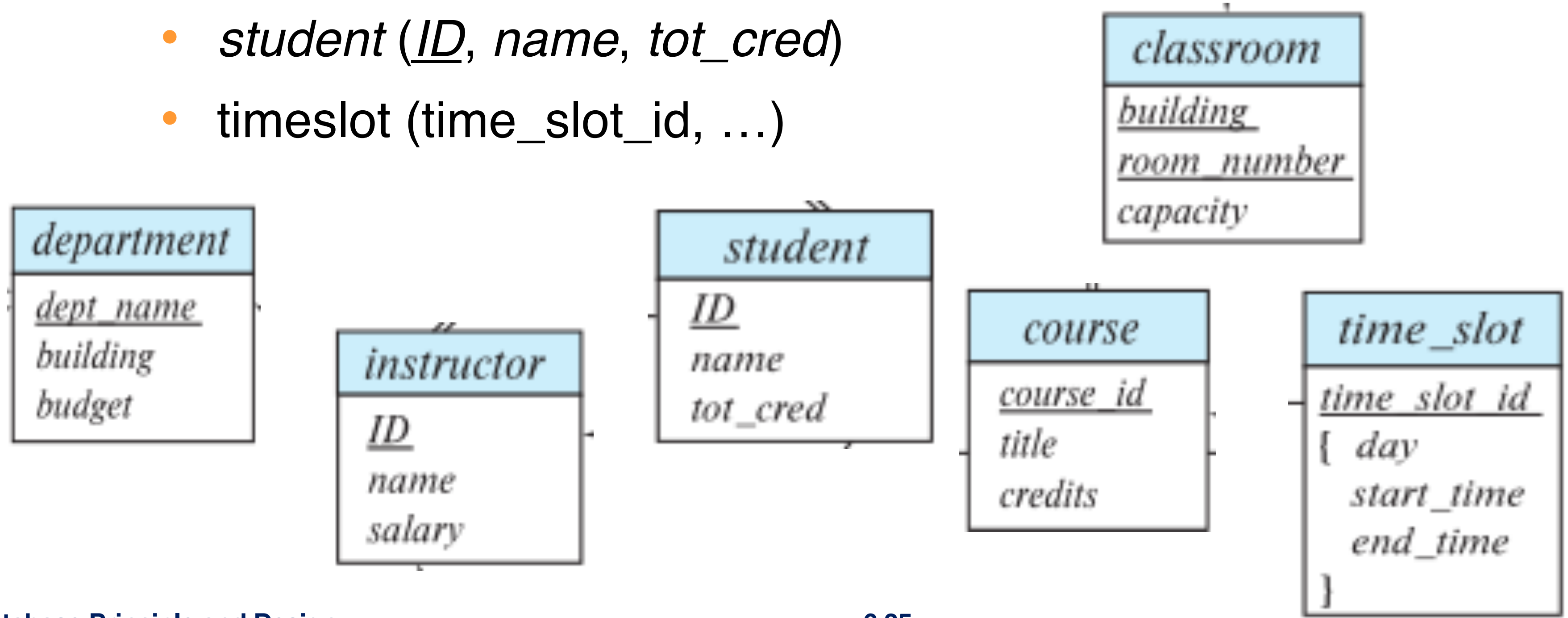
Note: A complex E-R diagram may need to be split into multiple pages, an entity set may appear in more than one page. The attributes of the entity set should be shown only once, in its first occurrence. Subsequent occurrences of the entity set should be shown without any attributes, to avoid repeating the same information at multiple places, which may lead to inconsistency.

6.7 Reducing E-R Diagram to Database Schema

Because the E-R model and the relational database model employ similar design principles, we can convert an E-R design into a relational design easily.

Representation of Strong Entity Sets

- A strong entity set reduces to a schema with the same attributes and the primary key.
- Schemas derived from strong entity sets in the university E-R diagram:
  - *classroom* (*building, room_number*, *capacity*)
  - *department* (*dept_name*, *building*, *budget*)
  - *course* (*course_id*, *title*, *credits*)
  - *instructor* (*ID*, *name*, *salary*)
  - *student* (*ID*, *name*, *tot_cred*)
  - timeslot (time_slot_id, …)



Representation of Weak Entity Sets

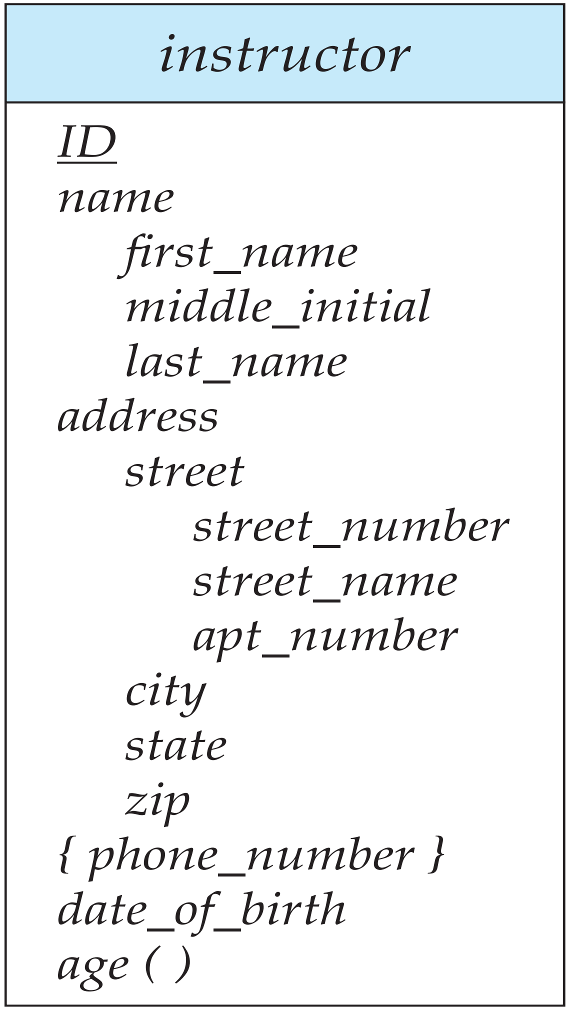
- Let *A* be a **weak entity set** with attributes *a*1, *a*2,…, *am*. Let *B* be the **strong entity set** on which *A* depends. Let the primary key of *B* consist of attributes *b*1, *b*2,…, *bn*.
- The relation schema R for A: {*a*1, *a*2,…, *am*} ∪ {*b*1, *b*2,…, *bn*}
- The primary key of R: the combination of the **primary key** of B and the **discriminator attributes** of A.
- The foreign-key of R: *b*1, *b*2,…, *bn*
- Example:

section (course_id, sec_id, sem, year)

Representation of Entity Sets with Composite Attributes

- Composite attributes are flattened out by creating a separate attribute for each component attribute
  - Example: Ignoring multivalued attributes, extended instructor schema is

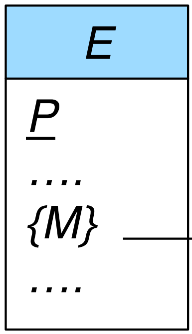
    instructor(ID,
    first_name, middle_initial, last_name,
    street_number, street_name,
    apt_number, city, state, zip_code,
    date_of_birth)



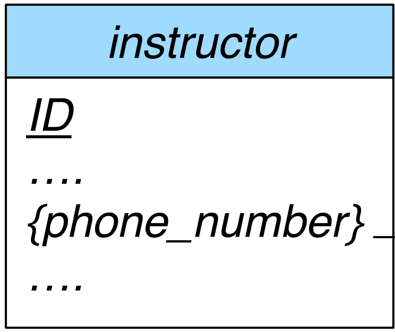
Note that a derived attribute is not explicitly represented in a relation schema. It may be represented in a view.

Representation of Entity Sets with Multivalued Attributes

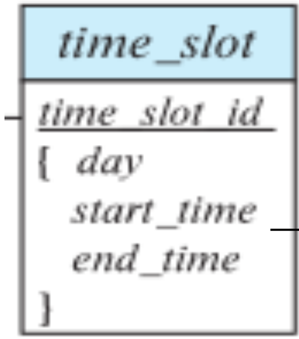
- A multivalued attribute *M* of an entity set *E* is represented by a separate schema *EM*
- Schema *EM* has attributes corresponding to the primary key of *E* and an attribute corresponding to multivalued attribute *M*
- The primary key of R: all attributes in EM
- Each value of the multivalued attribute maps to a separate tuple of the relation on schema *EM*
  - For example, an *instructor* entity with primary key 22222 and phone numbers 456-7890 and 123-4567 maps to two tuples: (22222, 456-7890) and (22222, 123-4567)



EM = (*P, M*)



inst_phone= (*ID, phone_number*)

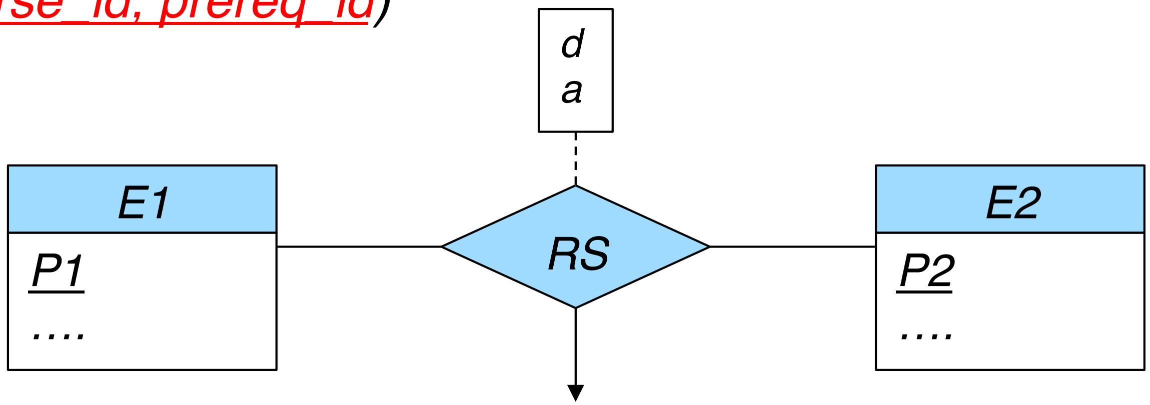


time_slot = (*time_slot_id, day, start_time, end_time*)

Representation of Relationship Sets

A many-to-many relationship set is represented as a schema R with attributes for the primary keys of two participating entity sets and all descriptive attributes of the relationship set.

- The primary key of R: attributes for the primary keys of two participating entity sets and discriminator attributes of the relationship set.
- Two foreign-keys of R: the primary key attributes of each participating entity set.
- Schemas derived from many-to-many relationship sets in the university E-R diagram:
  - *teaches (ID, course_id, sec_id, semester, year)*
  - *advisor (s_ID, i_ID) (actually advisor is a many-to-one relationship set)*
  - *takes (ID, course_id, sec_id, semester, year*, grade)
  - *prereq (course_id, prereq_id)*



$R = (\ \underline{P1, P2, d}, a\ )$

Representation of Relationship Sets (cont.)

- Many-to-one/one-to-many relationship sets can be represented by adding the primary key P1 of the “one” side and any descriptive attributes of the relationship set to the relation schema R of the “many” side. P1 is a foreign key of R.
- A foreign key is added to following schemas to represent following many-to-one/one-to-many relationship sets in the university E-R diagram:

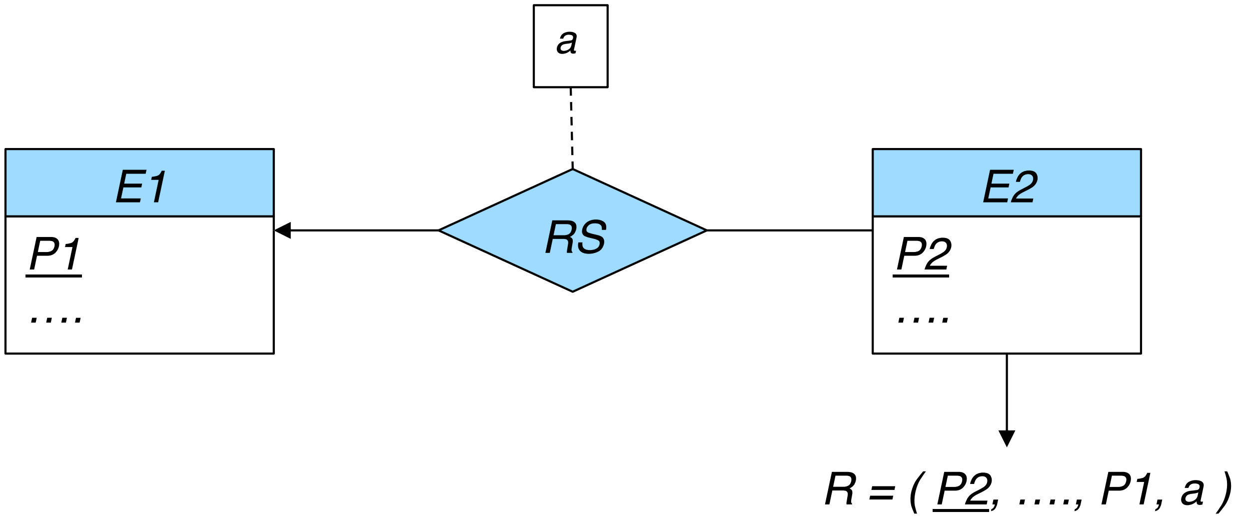
  inst_dept: *instructor (ID, name, salary, dept_name)*

  stud_dept: *student (ID, name, tot_cred, dept_name)*

  course_dept: *course (course_id, title, credits, dept_name)*

  sec_class: *section (course_id, sec_id, semester, year, building, room_number)*

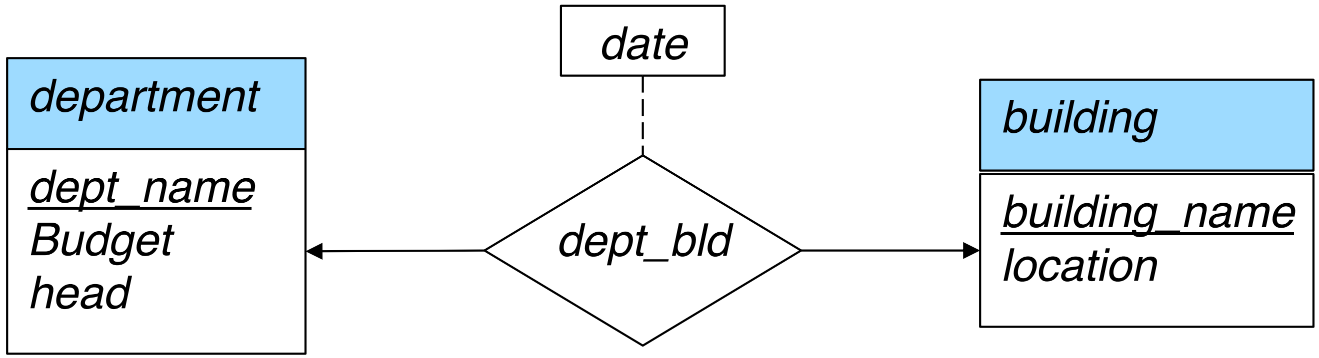
  sec_time_slot: *section (course_id, sec_id, semester, year, building, room_number, time_slot_id)*



Representation of Relationship Sets (cont.)

- One-to-one relationship sets can be represented by adding the primary key of one entity set and any descriptive attributes of the relationship set to the schema corresponding to another entity set.
- Example: the *dept_bld* relationship set can be represented either by
  - *department (dept_name, Budget, head, building_name, dept_bld_date)*

  or by
  - *building (building_name, location, dept_name, dept_bld_date)*



End of Chapter 6